



Cambridge International AS & A Level

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FURTHER MATHEMATICS**9231/22**

Paper 2 Further Pure Mathematics 2

October/November 2025**2 hours**

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.



- 1** The curve C has polar equation $r = e^{\frac{4}{3}\theta}$ for $0 \leq \theta \leq \pi$.

Find the length of C . Give your answer in an exact form.

[4]

[illegible]



- 2** Find the roots of the equation $z^4 = 8 - 8i\sqrt{3}$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi \leq \theta < \pi$. [5]

[illegible]



(b) Find the value of $\frac{d^2y}{dx^2}$ when $x = -1$.

[6]

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- 4 (a)** By considering the binomial expansion of $\left(z + \frac{1}{z}\right)^6$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\cos^6 \theta = a \cos 6\theta + b \cos 4\theta + c \cos 2\theta + d,$$

where a , b , c and d are constants to be determined. [5]

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5 (a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 5 \cos x. \quad [7]$$

[illegible]



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- (b) For large positive values of x and for any initial conditions, show that the solution to part (a) can be approximated by

$$y \approx R \sin(x + \phi),$$

where the constants R and ϕ are to be determined. [3]

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- 6 (a) Use the substitution $x = \frac{1}{2}\sqrt{2} \sinh u$ to find $\int \frac{1}{\sqrt{2x^2+1}} dx$. [3]

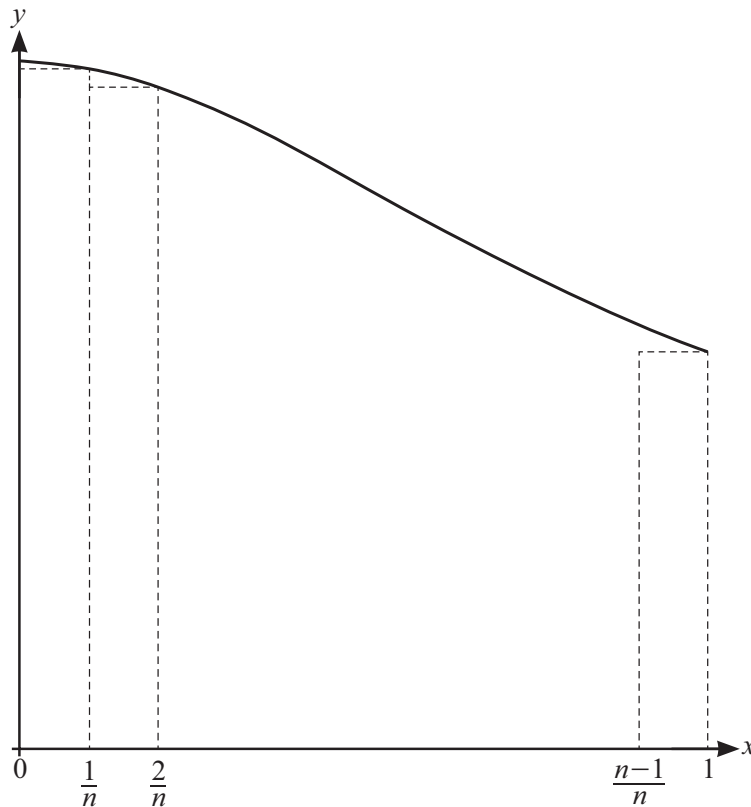
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The diagram shows the curve with equation $y = \frac{1}{\sqrt{2x^2+1}}$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

- (b) By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^n \frac{1}{\sqrt{2r^2+n^2}} < \frac{1}{2}\sqrt{2} \ln(\sqrt{2} + \sqrt{3}). \quad [5]$$

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- (c) Use a similar method to find, in terms of n , a lower bound for $\sum_{r=1}^n \frac{1}{\sqrt{2r^2 + n^2}}$. [4]

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- (d) Deduce the exact value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{2r^2 + n^2}}$. [1]

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7 (a) Show that $\frac{d}{dx}\left(\frac{1}{2}x\sqrt{4-x^2} + 2\sin^{-1}\left(\frac{1}{2}x\right)\right) = \sqrt{4-x^2}$. [3]

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(b) Find the solution of the differential equation

$$2\frac{dy}{dx} + \frac{y}{2+x} = 2\sqrt{2-x}$$

for which $y = \frac{1}{2}$ when $x = 1$. Give your answer in an exact form. [8]

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[illegible]

- 8 (a) Find the values of k for which the system of equations

$$x - y + 2kz = 1,$$

$$kx + y + 2z = 2,$$

$$2x - y + z = 3,$$

does not have a unique solution.

[3]

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- (b) Given that $k = -\frac{1}{2}$, show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically.

[3]

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- (c) Given instead that $k = -1$, show that the system of equations in part (a) is also inconsistent. Interpret this situation geometrically.

[4]

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The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{pmatrix}.$$

- (d) Use the characteristic equation of $\mathbf{A}$ to show that $\mathbf{A}^4 = p\mathbf{A}^2 + q\mathbf{A}$ where p and q are integers to be determined. [5]

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Additional page

If you use the following page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

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